Neddy on your watch board

Part 1: get your Waveshare ESP32-S3 AMOLED 2.06" board working on your Windows PC. No experience needed. Every step says what you should see. If something looks different, stop and send me a photo or screenshot; that is exactly what I need.

What this part does. It installs the free program that puts software on the board (Arduino IDE), then runs Waveshare's own test programs: first the screen, then the microphone and speaker. When those work, your board is healthy and ready for the Neddy software I am building. **Part 2** (loading Neddy itself and pairing it to your account) comes with the firmware, in the same style.

What you need

- Your Windows PC (Windows 10 or 11) with internet.
- The board, charged or plugged in.
- **A USB-C cable that carries data, not only charging.** A cable that came with a phone charger sometimes only charges. If the PC never notices the board, try another cable. This is the number one cause of problems.
- About 45 minutes the first time (most of it downloads).

A. Install the Arduino IDE

1. On your PC open a browser and go to `arduino.cc/en/software`.
2. Under **Arduino IDE 2.x** click **Windows · Win 10 and newer, 64 bits**. On the next page choose **Just download**.
3. Open the downloaded file (it is called something like `arduino-ide_2.x.x_Windows_64bit.exe`) and click **Next** / **I Agree** / **Install** through every screen, keeping the default choices. If Windows asks whether to allow it, click **Yes**. If it asks about installing drivers, click **Install**.

   > You should see: an Arduino IDE icon on the desktop. Open it. A window with a white code area opens.

B. Teach it about your board

4. In Arduino IDE click **File** › **Preferences**.

5. Find the box **Additional boards manager URLs**. Click in it and paste exactly this line:
   `https://espressif.github.io/arduino-esp32/package_esp32_index.json`
   Click **OK**.

6. On the left edge click the **Boards Manager** icon (the second icon, a small circuit board). In its search box type `esp32`.

7. Find **esp32 by Espressif Systems**. It has a version drop-down: choose **3.3.11** (Waveshare's examples were tested on that exact version) and click **Install**.

   > This is a big download (several hundred MB) and can sit for a few minutes. Wait until the output panel says *Installed*.

   > You should see: "INSTALLED" next to esp32 by Espressif Systems, version 3.3.11.

C. Get Waveshare's examples and libraries

8. In your browser go to `github.com/waveshareteam/ESP32-S3-Touch-AMOLED-2.06`. Click the green **Code** button, then **Download ZIP**.

9. Open your **Downloads** folder, right-click the ZIP, choose **Extract All...** then **Extract**.

   > You should see: a folder named `ESP32-S3-Touch-AMOLED-2.06-main`.

10. Open it, then go into `examples` › `arduino` › `libraries`. You will see folders such as `Arduino_GFX`, `lvgl`, `SensorLib`, `XPowersLib`, `Arduino_DriveBus` and `board_config`, plus a file `lv_conf.h`.

11. **Select everything inside that `libraries` folder** (click one item, press `Ctrl` + `A`) and copy it (`Ctrl` + `C`).

12 Open **File Explorer** and go to `Documents` › `Arduino` › `libraries`. (If there is no `libraries` folder inside `Arduino`, create it.) Paste (`Ctrl` + `V`). If Windows asks about replacing files, choose **Replace**.

> Use Waveshare's copies of these libraries, not ones from the Library Manager. Their examples need these exact versions (for example LVGL 9.3.0).

> You should see: folders `Arduino_GFX`, `lvgl`, `SensorLib`, `XPowersLib`, `Arduino_DriveBus`, `board_config` and the file `lv_conf.h` in `Documents\Arduino\libraries`.

13 **Close Arduino IDE completely and open it again** so it notices the new libraries.

D. Plug in the board and test the screen

14 Plug the USB-C cable into the board and into the PC.

> You should see: the board lights up and Windows makes its “device connected” sound. If the board seems off, press the **PWR** button once.

15 Right-click the Start button › **Device Manager** › open **Ports (COM & LPT)**.

> You should see: a new entry like **USB Serial Device (COM5)** or **USB JTAG/serial debug unit (COM5)**. The number may differ. Unplug the cable and plug it in again to watch which entry appears and disappears; that is your board.

> Nothing appears? Try a different USB cable first, then a different USB port. If it appears under **Other devices** with a yellow mark, tell me, and I will give you the driver step.

16 In Arduino IDE: **File** › **Open...** and choose `ESP32-S3-Touch-AMOLED-2.06-main\examples\arduino\examples\01_HelloWorld\01_HelloWorld.ino`

17 Set these in the **Tools** menu (this is the most important step; one wrong setting means a blank screen):

Tools menu item	Choose
Board	**ESP32 Arduino › ESP32S3 Dev Module**
Port	The COM port from step 15
USB CDC On Boot	**Enabled**
Flash Size	**32MB**
PSRAM	**OPI PSRAM**
Partition Scheme	one with large app space, for example **Huge APP** (for these tests any works; pick the biggest-app choice if the list has one)

18 Click the **Upload** button (the round arrow pointing right, top left). Wait. The first build can take a few minutes.

> You should see: the output panel scrolling, then **Done uploading**. The board screen shows a red **Hello World!** and then words at random places and colours.

> **If upload fails or the port is missing:** put the board in download mode. Hold the **BOOT** button, unplug and replug the USB cable while holding, then let go. Check the port again in the Tools menu and click Upload again. Afterwards, to run normally, unplug and replug (or press PWR). If the board turns off by itself, hold **PWR** for 6 seconds, then press it once.

E. Test the microphone and speaker

19 Open `examples\arduino\examples\08_ES8311\08_ES8311.ino` the same way (**File › Open...**). Keep the same Tools settings. Click **Upload**.

20 When it finishes, the board plays a built-in tune. Then it repeats what the microphone hears through the speaker.

> You should hear: a short piece of music, then your own voice played back when you talk. (This test does not use the screen, so the screen may stay blank.)

Open **Tools › Serial Monitor**, set the speed box to **115200**. You should see `[echo] Echo start`.

> No sound? Make sure a speaker is actually attached to the board (some kits ship the speaker separately; it plugs into the small 2-pin socket). Tell me what you see on the board and what the Serial Monitor says.

F. What to send me when you finish

- Did the screen test (step 18) work? A photo of the screen helps.
- Did you hear the music and your own voice (step 20)?
- Does the board have a speaker attached, and are the two side buttons labelled **BOOT** and **PWR**? A photo of the board from the front and side is great.
- Any step where what you saw was different from “You should see”. Screenshot the Arduino window, including the black output panel at the bottom.

If something goes wrong

What you see	What to do
No COM port appears	Different USB cable (data cable), different port; then BOOT-button download mode (step 18 warning).
Red errors mentioning `lvgl` or `Arduino_GFX` not found	Step 12 was not done or went to the wrong folder; redo it, then close and reopen Arduino IDE.
Screen stays black after “Done uploading”	Check Tools: PSRAM must be **OPI PSRAM**, USB CDC On Boot **Enabled**, Board **ESP32S3 Dev Module**. Unplug and replug.
Board turns itself off	Hold PWR about 6 seconds, then press PWR once. If a battery is attached and it keeps cutting out, disconnect the battery and try USB only.
Anything else	Screenshot the whole Arduino window and send it with a sentence on which step you were on.

These steps follow Waveshare’s own documentation for this board (docs.waveshare.com, ESP32-S3-Touch-AMOLED-2.06, Arduino chapter). I have not been able to try them on your PC, so tell me wherever a step does not match what you see, and I will fix this guide.